Unit: 6 Unit 8: Quadratics
Topic: Solving by Factoring and Other Methods

Name: ____________

Date: ________

Advanced (Habanero)

Q1. Solve $x^2 + 5x + 6 = 0$ by factoring:
(A) $x = -2, -3$
(B) $x = -1, -6$
(C) $x = 2, 3$
(D) $x = 1, 6$
(E) $x = -2, 3$

Q2. Solve $x^2 - 7x + 12 = 0$ by factoring:
(A) $x = 3, 4$
(B) $x = 5, 12$
(C) $x = 3, 12$
(D) $x = 4, 12$
(E) $x = 5, 3$

Q3. Which of the following is a solution to $x^2 - 4x - 5 = 0$?
(A) $x = -1$
(B) $x = 5$
(C) $x = -5$
(D) $x = 1$
(E) $x = 4$

Q4. Solve $x^2 + 2x - 15 = 0$ by factoring:
(A) $x = -5, 3$
(B) $x = 5, -3$
(C) $x = -3, 5$
(D) $x = -1, 15$
(E) $x = 1, -15$

Q5. Verify the solution $x = 2$ to $x^2 - 4x + 3 = 0$:
(A) True
(B) False
(C) Maybe
(D) Not enough info
(E) Unknown

Q6. Which equation has the solutions $x = -2, 3$?
(A) $x^2 + 5x + 6 = 0$
(B) $x^2 - 7x + 12 = 0$
(C) $x^2 - 1x - 6 = 0$
(D) $x^2 + 1x - 6 = 0$
(E) $x^2 - x - 6 = 0$

Q7. Solve $x^2 + x - 12 = 0$ by factoring:
(A) $x = -4, 3$
(B) $x = 4, -3$
(C) $x = 3, -4$
(D) $x = -3, 4$
(E) $x = 1, -12$

Q8. Solve $x^2 - 9x + 20 = 0$ by factoring:
(A) $x = 4, 5$
(B) $x = 5, 4$
(C) $x = 4, 9$
(D) $x = 5, 9$
(E) $x = 9, 4$

Open Ended Questions (FRQ)

Q9. Solve $x^2 + 8x + 15 = 0$ by factoring and verify the solutions.

Q10. Find all values of x for which $x^2 - 5x - 6 = 0$ and explain your method.

Chef's Tips

- Read each question carefully before answering.
- Use the back of the page for scratch work.
- Show all work and justify your answers.